

Classical Field Theory II

Lecture Notes

Symmetry breaking, topology, gravity, constraints, and effective viewpoints.



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Preface

Classical Field Theory II continues the first course after the basic Lagrangian, Hamiltonian, symmetry, and fundamental-field examples have been introduced. The emphasis now shifts toward nonlinear phenomena, topology, gravity, boundary charges, constraints, and the conceptual preparation for quantum field theory.

Remark

The guiding question of this second course is not only how to derive field equations, but how classical field theories organize phases, defects, geometry, gauge redundancy, boundaries, and effective descriptions.

Part I

Symmetry Breaking, Topology, and Nonlinear Fields

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Spontaneous Symmetry Breaking in Classical Field Theory

- 1.1 Symmetry of the Action vs Symmetry of the Vacuum
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